

Mahantesh Khetri

September 2026

Ph.D. Candidate, Electrical and Computer Engineering | University of Virginia

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[Website](#) | [Google Scholar](#) | [LinkedIn](#) | [Lab](#)

Research & Professional Experience

- 08/2022–Present ▶ **Graduate Research Assistant / Doctoral Researcher**
University of Virginia, Electrical and Computer Engineering
Advisor: Professor Mool C. Gupta
- Develop laser micromachining, cutting, patterning and surface-texturing processes using UV, visible and IR nanosecond, picosecond and continuous-wave systems; process sapphire, silicon and SiC substrates.
 - Develop selective silver-contact ablation and debonding routes for silicon and CIGS solar cells; demonstrated single-pass CIGS debonding at scan speeds up to **1,200 mm/s**.
 - Build image-to-toolpath workflows using Python, OpenCV and U-Net segmentation (**>98% pixel-level accuracy**); program Galil stage motion and configure EZCAD/galvo scan paths.
 - Develop a **355 nm Raman, fluorescence and LIBS platform** with common excitation/collection optics, fiber-coupled spectroscopy and motorized positioning.
 - Analyze **STEM-CL/HAADF** measurements of $\text{TiO}_x/\text{Al}_2\text{O}_3/\text{Si}$ interfaces and evaluate spectral-processing artifacts; perform hands-on **ellipsometry** and complementary materials metrology.
 - Investigate extreme-low-light RAW noise and restoration through repeated-measurement analysis and controlled U-Net/Restormer experiments on Sony and Fuji datasets.
 - Develop laser backsheet-delamination and thermal-modeling workflows; align and troubleshoot optical hardware, prepare SOPs and train laboratory users.
- 12/2019–07/2022 ▶ **Research Fellow** · *Indian Institute of Technology Delhi*
Department of Physics, New Delhi, India
- Conducted single-crystal growth and PLD thin-film research using high-vacuum systems; correlated structure, composition and electrical/electrochemical properties using microscopy, spectroscopy and PPMS measurements.
- 01/2022–07/2022 ▶ **PPMS Operator** · *Indian Institute of Technology Delhi*
Physical Property Measurement System, Department of Physics
- 07/2018–05/2019 ▶ **Research Student** · *Indian Institute of Technology Ropar*
Fabricated Sb-doped SnO_2 transparent conducting oxide films by thermal evaporation and controlled post-oxidation; characterized structural, optical and electrical properties.
- Summer 2017 ▶ **Research Intern** · *National Institute of Technology Warangal*

Education

- 2022–2026 ▶ **Ph.D., Electrical and Computer Engineering** · *University of Virginia*
Expected December 2026 · GPA: 3.9/4.0
- 2017–2019 ▶ **M.Sc., Physics** · *Indian Institute of Technology Ropar*
CGPA: 8.47/10
- 2013–2016 ▶ **B.Sc., Computer Science, Physics and Mathematics**
Rani Channamma University · 94.97% · University Second Rank; Gold Medal in Computer Science

Research Interests

Laser–material interactions and micromachining; optical instrumentation and spectroscopy; semiconductor–interface metrology; photovoltaic materials recovery; physics-informed computational imaging.

Technical Skills

Laser Processing	Laser cutting and micromachining; ablation; debonding; patterning; surface texturing; delamination; sintering; process-window development.
Optics & Spectroscopy	Nanosecond, picosecond and CW laser systems; UV/visible/IR optics; alignment and focusing; Raman; fluorescence; LIBS; ellipsometry; FTIR; UV-Vis.
Motion & Automation	Galil motion control; EZCAD; galvo scanners; F-theta optics; image-to-vector scan paths; Python/OpenCV machine vision; TTL/driver interfaces.
Materials & Devices	PLD; thermal evaporation; high-vacuum systems; single-crystal growth; silicon/CIGS photovoltaics; transparent conducting oxides; passivating contacts.
Characterization	SEM/EDS; TEM/STEM; cathodoluminescence; HAADF-correlated analysis; XRD; XPS; AFM; PPMS; I–V, carrier-lifetime and wettability measurements.
Computing & Analysis	Python; NumPy; pandas; SciPy; Matplotlib; OpenCV; CNN/U-Net segmentation; RAW-image analysis; thermal modeling; reproducible scientific workflows.

Research Publications

Journal Articles

- [1] **M. Khetri**, P. K. Kanaujia, and M. C. Gupta. “Laser removal of silicon solar cell backsheet while preserving tempered glass and silicon wafer,” *Solar Energy Materials and Solar Cells*, vol. 304, 114404, 2026. doi: [10.1016/j.solmat.2026.114404](https://doi.org/10.1016/j.solmat.2026.114404)
- [2] A. Chakraborty, **M. Khetri**, A. Surampudi, and M. C. Gupta. “Superhydrophilic and Superhydrophobic Surfaces with Reversible Wettability Using Laser Processing,” *ACS Omega*, vol. 11, no. 19, pp. 27970–27984, 2026. doi: [10.1021/acsomega.5c11755](https://doi.org/10.1021/acsomega.5c11755)
- [3] **M. Khetri**, P. K. Kanaujia, and M. C. Gupta. “Laser recycling of silver in bulk and nanoparticle form from silicon solar cells and deep learning for process automation,” *Renewable Energy*, vol. 254, 123684, 2025. doi: [10.1016/j.renene.2025.123684](https://doi.org/10.1016/j.renene.2025.123684)
- [4] A. Trivedi, **M. Khetri**, A. Surampudi, and M. C. Gupta. “Gold Metal Recovery from Electronic Waste through Laser Generation of Micro and Nanoparticles,” *ACS Omega*, vol. 10, no. 24, pp. 26033–26040, 2025. doi: [10.1021/acsomega.5c02781](https://doi.org/10.1021/acsomega.5c02781)
- [5] **M. Khetri** and M. C. Gupta. “Recycling of silver from CIGS solar cells using laser ablation and debonding,” *Solar Energy Materials and Solar Cells*, vol. 282, 113381, 2025. doi: [10.1016/j.solmat.2024.113381](https://doi.org/10.1016/j.solmat.2024.113381)
- [6] **M. Khetri** and M. C. Gupta. “Recycling of silver from silicon solar cells by laser debonding,” *Solar Energy*, vol. 270, 112381, 2024. doi: [10.1016/j.solener.2024.112381](https://doi.org/10.1016/j.solener.2024.112381)
- [7] P. Mudgal, H. Arora, J. Pati, M. K. Singh, **M. Khetri**, and R. S. Dhaka. “Electrochemical investigation of MoSeTe as an anode for sodium-ion batteries,” *Proceedings of the Indian National Science Academy*, vol. 88, no. 3, pp. 430–438, 2022. doi: [10.1007/s43538-022-00101-5](https://doi.org/10.1007/s43538-022-00101-5)

Patent & Technical Report

M. C. Gupta and M. Khetri. Use of Lasers to Selectively Remove Materials from Substrates. Published U.S. patent application **US20260008129A1**, January 8, 2026; PCT/US2023/069458. [Patent record](#)

M. C. Gupta and M. Khetri. High-Efficiency Recycled Silicon Solar Cell. University of Virginia / U.S. Department of Energy technical report, 2024. OSTI ID: 2588729. [Report record](#)

Conference Proceedings

- [1] **M. Khetri** and M. C. Gupta. "Machine Learning-Assisted Automated Laser Recovery of Silver Electrode Recovery from Waste Silicon Solar Cells," *IEEE Photovoltaic Specialists Conference (PVSC)*, 2025. doi: [10.1109/PVSC59419.2025.1113134](https://doi.org/10.1109/PVSC59419.2025.1113134)
- [2] M. C. Gupta, P. K. Kanaujia, **M. Khetri**, and A. Trivedi. "Laser recycling of photovoltaic and electronic waste," *SPIE Energy Harvesting and Storage: Materials, Devices, and Applications XV*, PC1345005, 2025. doi: [10.1117/12.3054074](https://doi.org/10.1117/12.3054074)
- [3] **M. Khetri** and M. C. Gupta. "Laser Recycling of Silver from Silicon and CIGS Solar Cells Through Bulk Laser Debonding and Nanoparticles by Laser Ablation," *IEEE Photovoltaic Specialists Conference (PVSC)*, 2024. doi: [10.1109/PVSC57443.2024.10748926](https://doi.org/10.1109/PVSC57443.2024.10748926)
- [4] **M. Khetri**, P. K. Kanaujia, and M. C. Gupta. "Laser Recycling of Silver from Waste Silicon Solar Cells," *IEEE Photovoltaic Specialists Conference (PVSC)*, 2023. doi: [10.1109/PVSC48320.2023.10360069](https://doi.org/10.1109/PVSC48320.2023.10360069)
- [5] **M. Khetri**, R. Ramarajan, N. P. Reddy, M. Kovendhan, K. V. A. Srivatsav, B. S. Rao, K. Thangaraju, and D. P. Joseph. "Stabilization of 5 wt % 'Sb' doped SnO₂ thin film by post oxidation of thermally evaporated metallic layer," *AIP Conference Proceedings*, vol. 2115, 030278, 2019. doi: [10.1063/1.5113117](https://doi.org/10.1063/1.5113117)

Manuscript Under Review

- [1] **M. Khetri**, A. Surampudi, and M. C. Gupta. Multimodal Raman, Fluorescence, and LIBS Measurements of Planetary Minerals Using a Single 355 nm Pulsed-Laser Source. *Under review, IEEE Journal of Selected Topics in Quantum Electronics*, 2026.

Manuscripts in Preparation

- [1] Separating robust from processing-dependent observables in STEM cathodoluminescence of TiO_x/Al₂O₃/Si passivating contacts. *In preparation for submission*, 2026.
- [2] Measured Noise Statistics and Learning Coordinates in Extreme-Low-Light RAW Restoration. *Manuscript in preparation*, 2026.
- [3] **M. Khetri** and M. C. Gupta. Laser sintering for p-n junction formation using doped silicon nanoparticles synthesized by laser ablation. *Manuscript in preparation*.

Teaching Experience

University of Virginia

Spring 2024 ▶ **Teaching Assistant** · *Electrical Engineering*
ECE-600r: Photovoltaics

Indian Institute of Technology Delhi

Spring 2022 ▶ **Teaching Assistant** · *Department of Physics*
BTech Third-Year Laboratory (PYP-22I&22)

2021 ▶ **Teaching Assistant** · *Department of Physics*
PYL-208: Photovoltaics · Spring and Fall semesters

Presentations

Oral & Invited Presentations

- July 2025 ▶ **Laser Generation of Silver Nanoparticles from Solar Cell Waste**
UPLAMP 2025, Summer School on Ultra-short Pulse Lasers Applications in Material Processing, Vilnius, Lithuania · July 7–11, 2025.
- June 2025 ▶ **Machine-Learning-Assisted Automated Laser Recovery of Silver Electrodes from Waste Silicon Solar Cells**
53rd IEEE Photovoltaic Specialists Conference (PVSC), Montréal, Canada · June 8–13, 2025.
- 2025 ▶ **Invited Oral Presentation · UVA Engineering Research Symposium**
Charlottesville, Virginia. Second place, Invited Speaker Presentation.
- June 2024 ▶ **Laser Recycling of Silver from Silicon and CIGS Solar Cells Through Bulk Laser Debonding and Nanoparticles by Laser Ablation**
52nd IEEE Photovoltaic Specialists Conference (PVSC), Seattle, Washington · June 9–14, 2024.
Finalist, Best Student Presentation Award (oral).
- May 2024 ▶ **Oral Presentation · ANGEL-7**
7th International Conference on Advanced Nanoparticle Generation and Excitation by Lasers in Liquids, Charlottesville, Virginia · May 26–31, 2024.
- June 2023 ▶ **Laser Recycling of Silver from Waste Silicon Solar Cells**
50th IEEE Photovoltaic Specialists Conference (PVSC), San Juan, Puerto Rico · June 11–16, 2023.
Finalist, Best Student Presentation Award (oral).

Poster Presentations

- 2025 ▶ **Laser-assisted silver recycling from waste solar cells: AI-driven economical approach**
UVA Engineering Research Symposium, Charlottesville, Virginia. First place, Electrical Engineering poster presentation.
- March 2024 ▶ **2024 SETO Peer Review**
U.S. Department of Energy, Solar Energy Technologies Office, Washington, D.C. · March 26–27, 2024.
- January 2023 ▶ **CHIPS and Science Act Workshop**
University of Virginia, Charlottesville, Virginia.
- December 2018 ▶ **Stabilization of 5 wt % Sb-doped SnO₂ thin film by post-oxidation of thermally evaporated metallic layer**
63rd DAE Solid State Physics Symposium, Hisar, Haryana, India · December 18–22, 2018.
Published in AIP Conference Proceedings, 2019.

Awards & Fellowships

- 2025–2026 ▶ **Wilson Bicentennial Graduate Fellowship**
University of Virginia. Competitive endowed fellowship awarded annually to one ECE graduate student for outstanding research.
- 2025 ▶ **First Place, Electrical Engineering Poster Presentation**
UVA Engineering Research Symposium. Recognized research on laser-assisted silver recovery and AI-driven process automation.
- 2025 ▶ **Second Place, Invited Speaker Presentation**
UVA Engineering Research Symposium. Recognized the invited oral presentation of doctoral research in laser processing and photovoltaics.
- 2025 ▶ **UPLAMP Travel Award**
Travel support to present research at the international summer school on ultrashort-pulse laser materials processing in Vilnius, Lithuania.
- 2023, 2024 ▶ **Best Student Presentation Award Finalist · IEEE PVSC**
Selected for the oral-presentation competition at the 50th and 52nd PVSC, recognizing research quality and the student's contribution.
- 2024 ▶ **NSF-Supported Travel Award · ANGEL-7**
Conference and tutorial support for participation in ANGEL-7, focused on laser-based nanoparticle generation and excitation in liquids.
- 2019–2022 ▶ **Institute Fellowship · IIT Delhi**
Institute-supported research fellowship for experimental physics and materials research in the Department of Physics.
- 2018 ▶ **Merit-cum-Means Scholarship · IIT Ropar**
Scholarship recognizing academic merit and providing financial support during the M.Sc. Physics program.
- 2016 ▶ **University Second Rank; Gold Medal in Computer Science**
Rani Channamma University. Graduated with 94.97%, ranked second in the university and recognized as the Computer Science subject topper.

Leadership & Service

- 2018–2019 ▶ **Secretary and Social Media / External Affairs Lead**
Pehchaan Ek Safar, IIT Ropar. Founding member of the education and skill-development initiative for underserved children.
- 2015–2016 ▶ **Sports Secretary · BLDEA's College, Jamakhandi**

Mentoring and laboratory training: Participant in the UVA Graduate School Mentor Program; train laboratory users in laser operation, experimental procedures and safety.

References

Professor Mool C. Gupta

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Industry references are available upon request.